Hypothesis testing CHEAT SHEET

HYPOTHESES AND THE STEPS

research hypothesis → H_a: redesign lifts orders $\mu > 52$

challenge a claim → claim is H_0: cells $\mu \geq 500$

decision → H_0 is the target: rods $\mu = 250$ mm

lower tail $H_0\ \mu \geq \mu_0$, $H_a\ \mu < \mu_0$ · upper $H_0\ \mu \leq \mu_0$, $H_a\ \mu > \mu_0$

two-tailed $H_0\ \mu = \mu_0$, $H_a\ \mu \neq \mu_0$ · the equality is always in H_0

steps: hypotheses → α → statistic → p ≤ α ? → what it means

choose the tail(s) before seeing the data

TYPE I AND TYPE II ERRORS

Type I: reject a true H_0, $P = \alpha$ (level of significance)

Type II: do not reject a false H_0, $P = \beta$; power $= 1 - \beta$

say "do not reject H_0", never "accept H_0"

smaller α → larger β; a larger n lowers both

READING A P-VALUE

p = P(a result this extreme | H_0), not P(H_0 is true)

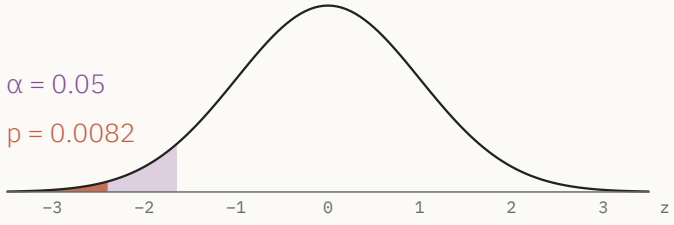
small p: strong evidence against H_0, not proof of H_a

p ≤ α → reject; report p itself, with an interval

eight cups, all right by luck: $1/70 \approx 0.014$

six sigma ≈ 1 in a billion by chance; a faulty cable isn't chance

ONE MEAN, SIGMA KNOWN: Z



$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ cells: $\frac{488 - 500}{40/\sqrt{64}} = \frac{-12}{5} = -2.4$

$p = P(Z \leq -2.4) = 0.0082 \leq 0.05$ → reject H_0

critical: $z \leq -z_\alpha = -1.645$ ($\bar{x} \leq 491.8$ cycles)

two-tailed: $p = 2\,P(Z \geq |z|)$; reject if $|z| \geq z_{\alpha/2} = 1.96$

rods: $z = \frac{0.18}{0.4/\sqrt{16}} = 1.8$, $p = 0.072$ → do not reject

interval $\bar{x} \pm 1.96\,\sigma/\sqrt{n} = 249.984 \ldots 250.376$, 250 inside

a two-tailed test at α rejects exactly the μ_0 outside the $1 - \alpha$ confidence interval

ONE MEAN, SIGMA UNKNOWN: T

$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$, $df = n - 1$ invoices: $\frac{33.2 - 30}{8/\sqrt{25}} = 2.0$

$p = \text{T.DIST.RT}(2, 24) = 0.0285$; critical $t_{0.05,\,24} = 1.711$

fatter tails than z; → z as n grows; small n: check skew

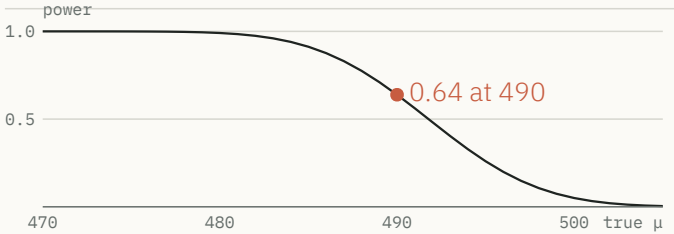
ONE PROPORTION

$z = \frac{\bar{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$ trials: $\frac{0.34 - 0.30}{0.0229} \approx 1.75$

upper p ≈ 0.040 → reject; use p_0 (not $\bar{p}$) in the SE

normal ok if $np_0 \geq 5$ and $n(1 - p_0) \geq 5$ (120, 280)

POWER, SAMPLE SIZE, SIGNIFICANCE



β at $\mu = 490$: $P(\bar{x} > 491.8) = 0.361$, power 0.639

at $\mu = 485$: power 0.912; at μ_0 the power equals α

$n = \frac{(z\alpha + z\beta)^2\,\sigma^2}{(\mu_0 - \mu_a)^2} = \frac{2.927^2 \times 1600}{100} \approx 137 \rightarrow 138$

two-tailed: $z_{\alpha/2}$ in place of z_α; always round n up

huge n: 4.62 vs 4.60 min, 500,000 users each

$z = 5$, $p < 10^{-6}$, yet the gap is under 2 seconds

report an interval with the p-value:

significant is not the same as important

Hypothesis testing CHEAT SHEET

TWO MEANS, SIGMA KNOWN

$z = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}}$ warehouses: $\frac{1.1}{0.474} \approx 2.32$

$p \approx 0.020$; interval $1.1 \pm 1.96 \times 0.474 = 0.17 \ldots 2.03$ min

TWO MEANS, SIGMA UNKNOWN: WELCH

$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(s_1^2/n_1 + s_2^2/n_2)}}$ pages: $\frac{60 - 52}{3.60} \approx 2.22$

$df = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)} \approx 23.8$

$p \approx 0.036$ = T.TEST(old, new, 2, 3)

by hand, round the df down (23)

pooled t (type 2) assumes one variance:
avoid it when the spreads differ

WHICH TEST?

one mean: σ known → z; σ unknown → t (df $n - 1$)

one proportion → z with p_0 in the standard error

two independent means → Welch's t (T.TEST type 3)

the same units measured twice → t on d (type 1)

two proportions → pooled z; the interval unpooled

A/B test → two proportions, n fixed in advance

MATCHED SAMPLES

work on $d = x_1 - x_2$: $t = \frac{\bar{d}}{sd/\sqrt{n}}$, $df = n - 1$

servers d = 11, 6, 7, 14, 4, 13, 6, 9: $\bar{d} = 8.75$, $s_d \approx 3.62$

as independent samples: $p \approx 0.36$ (the pairing matters)

$t \approx 6.8$, $p \approx 0.0002$ = T.TEST(…, 2, 1)

TWO PROPORTIONS AND A/B TESTS

test: pooled $\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$, $z = \frac{\bar{p}_1 - \bar{p}_2}{\sqrt{(\bar{p}(1 - \bar{p})(1/n_1 + 1/n_2))}}$

6,000 a page: 4.0 vs 4.6 %, $\bar{p} = 0.043$
$z \approx 1.62$, $p \approx 0.105$: could be luck

20,000 a page, same rates: $z \approx 2.96$, $p \approx 0.003$

interval: unpooled SE; lift 0.2 … 1.0 points

plan: $n = \frac{(z\alpha/2 + z\beta)^2 (p_1q_1 + p_2q_2)}{(p_1 - p_2)^2} \approx 17{,}940$ a page

fix n first; no peeking, no unplanned segments

CASE: WRENMOOR ENERGY (FICTIONAL)

400 homes, 200 given the display at random

kWh a month: a year before, and in the trial

after only: $p \approx 0.44$ (the spread hides it)

display homes only: −78.8 (the weather too)

the changes: saving 30.35 kWh, $t \approx -4.91$, $p < 10^{-6}$

95 %: 18.2 … 42.5 kWh, above the 15 kWh bar

roll out: payback ≈ 10 months (18.7 at the low end)

SPREADSHEET AND PYTHON

NORM.S.DIST(z, TRUE) · NORM.S.INV(p)

T.DIST(t, df, TRUE) · T.DIST.RT · T.DIST.2T

T.INV(p, df) · T.INV.2T(α, df) · ROUNDUP

T.TEST(r1, r2, tails, type):
1 matched · 2 pooled · 3 Welch

AVERAGE · STDEV.S · COUNT · SQRT

AVERAGEIF · COUNTIF · COUNTIFS

Python (scipy.stats): norm, t, ttest_1samp,
ttest_ind(…, equal_var=False), ttest_rel